

# **Lost in Translation: An Autonomous AI and Language Translation Study**

An Applied Research White Paper Examining Translation Accuracy, Autonomous Translation Risk, LLM Autonomous Evaluation, and the Need for Human Review

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## Abstract

As Large Language Models become embedded in global communication, education, and organizational operations, the question of whether these tools translate languages accurately has significant real-world consequences—for linguistic sovereignty and for how cultures, identities, and knowledge are represented within AI systems and digital ecosystems. This applied research white paper presents a replicable, rubric-based framework for evaluating LLM translation quality across a high-resource and a low-resource language using native speaker review to assess cultural and linguistic accuracy, a Forward and Reverse Translation Audit conducted by an IEEE CertifAIED™ Lead Assessor to detect Bias Laundering and Linguistic Masking, and a five-prompt LLM autonomous evaluation sequence to determine whether models can accurately predict and assess their own translation capabilities.

Six Large Language Models were evaluated across two types — five commercial/proprietary and one open-source/sovereign. Findings reveal significant performance variation across models and languages. No model excelled across all evaluation dimensions in either language. The only open-source model evaluated in this study scored lowest in both language translations.

The Forward and Reverse Translation Audit confirmed Bias Laundering: systematic bias introduced in the forward translation and masked in the reverse translation, invisible to English-only readers. The five-prompt LLM autonomous evaluation sequence revealed a consistent and significant gap between model autonomous assessment and actual performance.

This paper establishes the case for Human-in-the-Loop evaluation as a requirement in language translation deployments — particularly in critical, global, and organizational workflows where accuracy, cultural integrity, and linguistic sovereignty are not optional.

# 1. Introduction

## 1.1 The Ethical Imperative of Multilingual AI

The rapid proliferation of Large Language Models has transformed how organizations communicate, translate, and deliver services across languages and cultures. Proprietary Large Language Models have been built on training data that is predominantly English language centric, with non-English languages represented in varying and often insufficient proportions (Stanford HAI, 2025). The only open-source model in this study, Apertvs, states they support over 1,000 languages and long context, uses only fully compliant and open training data, and achieves comparable performance to models trained behind closed doors (Hugging Face, 2025).

This study begins from a simple premise: accurate translation is a fundamental requirement of ethical, responsible, and transparent AI deployment, with real-world impact on the sovereignty of languages and the future of global communication.

## 1.2 Origin of This Study

This study originated from the Women in AI 2025 Call for Experts Speaker Series, an initiative developed by Karen Jensen in her role as Senior AI Ethics Architect on the Global Ethics & Culture Team at Women in AI. During the series, expert Ishween Kaur presented “Intersectional Bias: Localization Challenges in AI and Solutions” (WomenInAI.co, 2025). The concepts explored during that session highlighted localization as an emerging AI ethics challenge and inspired the development of the present applied research study.

## 1.3 Research Objectives

- Evaluate the extent to which proprietary and open-source Large Language Models demonstrate culturally and linguistically appropriate translation quality across high and low-resource languages (Laumann, 2022).
- Assess whether Apertvs (the only open-source model), trained in a wider variety of languages with broader resource attribution — demonstrates measurable translation advantages over proprietary/commercial models.
- Determine what variations in language translation categories, per model performance, reveals about tool selection for culturally complex, ethically responsible translation tasks.
- Examine whether proprietary/commercial and open-source Large Language Models mask bias introduced during English-to-target-language translation through the reverse translation process — and whether English-only readers can detect that linguistic masking.
- Determine whether Large Language Models can autonomously and accurately translate languages, predict and assess their own translation capabilities, identify failures when asked to evaluate themselves, and recognize their own outputs.

## **1.4 Study Approach: LLM Tool Selection Perspective**

Organizations selecting AI translation tools may rely on factors such as reputation, cost, familiarity, or vendor claims without independently validating translation quality in the languages relevant to their use cases. This study proposes, and applies, a pre-defined, rubric-based, human-centered evaluation methodology that helps organizations evaluate whether a translation tool performs effectively for their specific language translation needs.

## **2. Ideas and Concepts Defined**

### **2.1 Defining Language Translation**

Language translation is the process of converting the written word from one language into another in a way that is culturally and linguistically appropriate so it can be understood by its intended audience (Markert, 2026).

### **2.2 High-Resource and Low-Resource Languages**

The terms high-resource and low-resource originate in Natural Language Processing (NLP) and refer not to the number of speakers a language has, but to the volume of digital data available to train language models. High-resource languages are characterized by large datasets — billions of written sentences from the internet, social media, and digital archives — giving NLP systems sufficient training data to perform accurately. Low-resource languages lack this data infrastructure, regardless of how many people speak them (Laumann, 2022).

This distinction explains a counterintuitive reality: Hindi, spoken by more than 600 million people, is classified as a low-resource language, while French, spoken by approximately 100 million people, is classified as a high-resource language. The difference is not population — it is digital representation, which has historically favored English and Western European languages (Laumann, 2022).

Each language type has inherent translation risks. High-resource languages risk homogenization — models' default to standard forms, flattening regional dialects and cultural variation. Low-resource languages risk English interference — models trained predominantly on English data produce output that reflects English reasoning patterns, resulting in unnatural phrasing, code-mixing, or loss of linguistic authenticity (Pava et al., 2025).

### **2.3 Human-in-the-Loop (HITL) Evaluation**

The Human-in-the-Loop approach integrates human expert review into AI-driven processes to ensure accuracy, cultural integrity, and ethical compliance (Kouahla, 2025).

### **2.4 Methodological Exclusion of Automated Translation Metrics**

In NLP research, BLEU, ROUGE, and ChrF are automated metrics designed to evaluate machine-generated translations by comparing them against a human-provided reference translation. They measure lexical overlap — how closely the output matches the reference — not meaning, cultural appropriateness, or ethical framing. The limitations of these metrics for evaluating translation quality in low-resource and culturally complex language contexts are well documented (Pava et al., 2025).

This study did not produce reference translations and did not apply automated metrics, so these NLP methodologies were not applied. Native speaker expert review was determined to be the more valid and appropriate methodology for evaluating culturally complex translation tasks.

## **2.5 Intersectional Bias in LLM Language Translation**

This study evaluates intersectional bias in translation as the compounded effect that occurs when singular biases — gender, language resource level, and cultural context — interact within multilingual AI systems, producing risks greater than any single bias alone. These factors may create compounded translation risks that are not readily detected through automated evaluation metrics or monolingual review. The Cultural Nuance and Resonance dimension of the evaluation rubric operationalizes this framework by assessing whether translations preserve, alter, or distort the gendered, hierarchical, and cultural elements of the source content.

## 3. Methodology

### 3.1 Source Document

The source document is the Women in AI blog post “2025 Women in AI Expert Series: Solving the Digital Divide with Artificial Intelligence” (Women in AI, 2025).

As a single source document, findings reflect performance on this specific text type and may not generalize to other content categories.

This source document was selected for three reasons: it contains technical AI terminology that presents a consistent translation challenge across all models; it includes professional titles and named expert contributors that allow evaluation of gendered language handling; and it represents the type of organizational content Women in AI produces for a global audience — making translation quality directly relevant to the organization’s mission.

### 3.2 Models Tested

| Type                     | Model                          |
|--------------------------|--------------------------------|
| Commercial / Proprietary | Gemini Flash 3                 |
| Commercial / Proprietary | ChatGPT 5.3                    |
| Commercial / Proprietary | ChatGPT 5.4                    |
| Commercial / Proprietary | ChatGPT 5.5                    |
| Commercial / Proprietary | Claude Sonnet 4.6              |
| Commercial / Proprietary | Perplexity.ai — Model Agnostic |
| Open Source / Sovereign  | Apertvs-8B-Instruct-2509       |

*No preference for or endorsement of any specific model or platform is expressed or implied.*

### 3.3 Language Selection

| Language | Family & Description                                                                                                                                                                                                                                                                     | Resource Type | Reviewer      | Review Status                    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|----------------------------------|
| Mandarin | Sino-Tibetan (Britannica, 2026). One of the world’s largest language families, spanning China, Southeast Asia, and the Himalayan region. Tonal, isolating language — words do not change form for tense, gender, or number. Evaluated against Standard Putonghua. Risk: homogenization — | High Resource | Charlotte Tao | Evaluation Completed May 6, 2026 |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |              |                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|-----------------------------------|
|       | models’ default to Putonghua, flattening regional dialects and cultural variation.                                                                                                                                                                                                                                                                                                                                                                                                    |              |              |                                   |
| Hindi | Indo-European: Indo-Aryan (Britannica, n.d.). Indo-European is the world’s most widely distributed language family. Written in Devanagari script. Has grammatical gender — nouns, verbs, and adjectives change form based on gender, making gendered language assumptions a critical evaluation dimension. Evaluated against Manak Hindi (Standard Modern Hindi). Risk: English interference — models produce output reflecting English patterns, unnatural phrasing, or code-mixing. | Low Resource | Ishween Kaur | Evaluation Completed June 7, 2026 |

### 3.4 Translation Input Methodology

All translations were produced using a single standardized prompt with no additional context, examples, or iterative refinement. This technique assesses each model's intrinsic translation knowledge rather than the quality of the prompt or the skill of the operator.

- Forward translation prompt: “Please translate the following into [language]. At the end of the translation, identify and summarize any modifications you made for cultural nuances and/or to mitigate bias.”
- Reverse translation prompt: “Please translate this to American English.”

Cultural adaptation summaries provided by each model at the end of the forward translation were retained for research analysis (see Section 3.9) but were not included in reviewer files to prevent anchoring bias in scoring.

### 3.5 Blind Review Protocol

Mandarin (Charlotte Tao): ZH-C (ChatGPT 5.4) and ZH-E (Gemini Flash 3) are declared non-blind — Charlotte has prior copies of these outputs. These two files are clearly identified by model name in the reviewer document header. The remaining four models are presented under full blind conditions.

Hindi (Ishween Kaur): No prior exposure to any of the six model outputs. All six files presented under full blind conditions — the strongest possible blind review position in the study, and a methodological contrast to the Mandarin partial blind protocol.

## 3.6 Evaluation Rubric

Translations were scored by native speaker reviewers on a 1–5 scale across five dimensions, where 5 represents the highest quality and 1 represents significant deficiency. Dimensions were scored independently.

| Weight | Dimension                         | Definition                                                                                                                                             |
|--------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20%    | Semantic Fidelity                 | Meaning consistency rather than word-for-word equivalence (Semantic Fidelity, n.d.)                                                                    |
| 25%    | Linguistic Accuracy / Sovereignty | Grammar, vocabulary, and structure correct in the target language; resistance to drift toward dominant languages (The Importance of Accuracy..., 2023) |
| 20%    | Cultural Nuance / Resonance       | The invisible architecture of meaning — humor, etiquette, social hierarchies, gendered language (Chiorescu, 2025)                                      |
| 15%    | Terminology Precision             | Correct and consistent use of specialized AI and technology terminology (Bergner, n.d.)                                                                |
| 20%    | Localization Fluency              | Does it read as native, not translated? Standard Putonghua for Mandarin; Manak Hindi for Hindi (Owens-Hughes, 2025)                                    |

Weighted composite:  $(SF \times 0.20) + (LA \times 0.25) + (CN \times 0.20) + (TP \times 0.15) + (LF \times 0.20)$

### 3.6.1 Rubric Dimension Definitions

#### Localization Fluency

Translation converts words. Localization converts meaning. While this study uses translation as its primary test methodology, localization — the broader process of ensuring cultural relevance, preference, and appropriateness — is the standard against which translation quality is ultimately measured (Owens-Hughes, 2025).

#### Cultural Nuance / Resonance

Cultural nuance is the invisible architecture supporting meaning — humor, emotions, etiquette, metaphors, social hierarchies, and the unspoken assumptions behind words (Chiorescu, 2025).

#### Terminology Precision

Terminology precision refers to the use of specialized language appropriate to a particular subject or context (Bergner, n.d.). In this study, AI terms with particular evaluation significance include: Prompt Engineering, Agentic AI, Large Language Model, Digital Divide, and NSFW (Not Safe for Work).

#### Linguistic Accuracy / Sovereignty

Linguistic accuracy refers to how correctly a model applies the grammatical, phonological, and lexical rules of the target language (The Importance of Accuracy..., 2023). Linguistic sovereignty

extends this to encompass a language’s resistance to being overwritten by dominant neighboring languages or training data biases.

### **Semantic Fidelity**

Semantic fidelity focuses on meaning consistency rather than word-for-word equivalence (Semantic Fidelity, n.d.). A translation that is grammatically correct but misrepresents the meaning of the source fails at the most fundamental level.

### **Confidence**

Language reviewers were asked to rank how certain they were in their own scores on a scale of 1–5 where 5 represents complete confidence. Confidence ratings are noted alongside reviewer scores in Sections 4.1 and 4.2.

## **3.7 Forward and Reverse Translation Audit**

### **Linguistic Masking and Bias Laundering**

Linguistic Masking describes a phenomenon in this study where a model generates biased or gendered output in the target language that is invisible to English-only auditors (Ligteringen, n.d.). While there is no single origin of the term Bias Laundering, in this paper we’ve identified Bias Laundering as a systematic translation-reverse translation phenomenon: a model introduces biased language in the English to target language, then restores the appearance of neutrality in the reverse translation of target language to English (New York State Unified Court System, 2025).

Each model translated the source document from English into the target language (forward translation), then translated its own output back to American English (reverse translation). The resulting reverse translation (target language to English) is compared against the original source to identify Bias Laundering and Linguistic Masking — phenomena in which gendered or hierarchical assumptions introduced in the target language are masked from English-only readers.

All six models performed their own reverse translation. Same-model reverse translation is the scenario most likely to produce Bias Laundering — the model may smooth over assumptions it introduced in the forward translation.

The Forward and Reverse Translation methodology was developed specifically for this study. The audit was conducted by Vanessa Maidoh, IEEE CertifAIEd™ Lead Assessor, under IEEE 7003-2024: Standard for Addressing Algorithmic Bias Considerations in Autonomous and Intelligent Systems. Her credential establishes expertise and authority in assessing the ethicality of AI-based products and systems — including Algorithmic Bias, Transparency, Accountability, and Privacy.

Bias Laundering and Linguistic Masking findings were derived through comparative analysis of the original English source text, forward translations, reverse translations, and reviewer observations.

To support consistent identification and classification of Bias Laundering and Linguistic Masking findings, the study applied the following classification framework:

### **Classification Framework (IEEE 7003-2024):**

- COMPLIANT — reverse translation matches source in meaning, terminology, and register
- DRIFT PRONE — visible difference between source and reverse detectable by English reader
- MASKED (Bias Laundering) — forward translation contains a bias, distortion, or terminology change that is corrected or normalized in the reverse translation, making the issue difficult for an English-only reviewer to detect.
- CONTENT OMISSION — sentence, phrase, or required element entirely absent from reverse
- AUDIT LIMITATION — assessment required the combined use of forward translation analysis, reverse translation analysis, and native-language reviewer observations to reach a determination.

### 3.8 A Note on Model Version Evolution in Emerging AI Research

A persistent challenge in conducting comparative research on proprietary/commercial AI systems is their dynamic versioning. While open-source models like Apertvs are released as fixed, fully documented checkpoints with accessible changelogs (Hugging Face, 2025), proprietary/commercial LLMs are fluid, dynamic systems. These models are continuously updated, fine-tuned, and redeployed under the same product name without a formal changelog accessible to researchers or end users.

This study encountered this challenge directly. During the Mandarin run, ChatGPT self-identified as version 5.3. OpenAI’s model release notes confirm that as of March 2026, GPT-5.3 Instant served as the default ChatGPT model, with GPT-5.4 designated for Thinking and Pro modes (OpenAI, 2026a). Approximately 45 days later, during the Hindi run, GPT-5.5 had been released (OpenAI, 2026b) and was actively replacing GPT-5.3 as the default. The same model label consequently presented as version 5.3–5.5 Instant (mostly 5.5, GPT self-reported) and a separate 5.4 Thinking mode — capturing a major version transition in progress.

This illustrates a structural feature of proprietary/commercial LLM research: designations such as “Instant,” “Thinking,” and “Pro” are operating modes, not fixed model versions, and the underlying version each mode maps to changes as new releases roll out. ChatGPT additionally auto-routes between Instant and Thinking based on task complexity (OpenAI, 2026a), meaning a single recorded session may not correspond to a single stable model configuration. A model version recorded at the time of testing therefore represents a timestamped snapshot of a continually evolving system rather than a durable identifier. As a result, study findings should be interpreted within the context of the model versions available at the time of evaluation.

### 3.9 LLM Autonomous-Reported Translation Processes

Each model was prompted to identify and summarize any modifications made for cultural nuances and/or bias mitigation at the end of the forward translation. The analytical value of these summaries lies in comparing what each model claimed it achieved, against what native-speaker reviewers actually scored. The gap between model autonomous-assessment and human evaluation forms a core empirical finding of this study (see Section 4.5 and Section 6). To maintain

a strict double-blind evaluation environment, these bias identifications and self-reports were completely removed from the data files provided to the reviewers.

### **3.10 LLM Autonomous Audit Protocol**

Each model was given the study's results — reviewer scores, file codes, and the models' bias summaries, all identified only by code — and asked to analyze them as an independent evaluator, without being told that one of the files was its own output. Across a five-prompt sequence, each model assessed its own anticipated capability, evaluated the blind results, attempted to identify which file was its own, and was then shown the answer. Because the model did not know it was judging its own work, the sequence functioned simultaneously as a peer review of the full set and an unwitting autonomous audit.

The five prompts were run sequentially, not combined, to prevent earlier responses from contaminating later ones: (1) capability autonomous-assessment and predicted scores; (2) blind results presentation; (3) bias summary evaluation; (4) autonomous identification; and (5) reveal and reflection.

ChatGPT 5.5 Instant and Gemini Flash 3 ran the five-prompt sequence for Mandarin; Apertvs and Perplexity.ai ran it for Hindi. The prompt sequence was assessed and refined using a neutral WAI blog post excerpt before being run on study subjects. Findings are documented in Section 4.5.

### **3.11 Study Validation: Expert Persona Prompting**

Throughout this study, every attempt has been made to incorporate, use and deploy existing methodologies for interactions with LLMs. A common methodology, especially for research and engineering, is to use Expert Persona Prompting (EPP). EPP defines the viewpoint of an expert, in this case, a PhD applied researcher, to evaluate the study and provide feedback (EmergentMind, 2025).

This paper's concept was originally theorized as pure research. But due to the dynamic nature of the LLM models and the conditions of emerging technology research, the study evolved into an Applied Research White Paper. The intent in reviewing study content using EPP was to identify any research failures and/or fallacies that needed to be addressed.

The EPP was developed in Gemini Flash 3 (free tier) with the PhD Applied Researcher persona. The study was then submitted to the EPP for review.

The EPP review produced five methodological observations:

- Research classification — whether the study met the standard for applied research
- Rubric design — whether the five dimensions and weightings were sufficiently justified
- Source document scope — whether a single source document was adequate for comparative analysis
- Blind review protocol — whether partial blind conditions in Mandarin compromised evaluation integrity

- Automated metric exclusion — whether the decision not to apply BLEU, ROUGE, or ChrF required stronger methodological justification

Upon review, the current study had already addressed each observation within its existing methodology and framing. No structural changes to the study were required as a result of the EPP review.

### 3.12 LLM Outputs on Interpreting ‘American English’

Three models were asked “*What is your understanding of American English in this context and how did you determine how to execute that understanding?*”:

- ChatGPT 5.3-5.5 Instant provided a five-part framework covering lexical choices, tone and register, syntax and flow, technical terminology normalization, and cultural framing — with specific examples for each. It also documented what it deliberately did not do, including avoiding over-literal translation and marketing register.
- Gemini Flash 3 grounded its response in training sources — referencing AP Stylebook, Chicago Manual of Style, APA, and MLA as style authorities, and Merriam-Webster vs. Oxford as the dictionary authority. It noted pattern recognition from American vs. British sources and ended by asking whether the output matched the “flavor” of English requested.
- Claude Sonnet 4.6 interpreted the prompt through genre — identifying the source as a professional NGO/advocacy document — and provided specific examples of translation choices made: “break into emerging tech,” “deep dive,” “upskill,” and “takeaway” as a section header. It also documented what it left unchanged.

### 3.13 Reference Selection

Because this study examines a rapidly evolving area of AI practice, the literature includes a combination of peer-reviewed research, technical standards, industry documentation, organizational publications, and proprietary/commercial model documentation. Sources were selected based on their relevance to the research questions, methodological transparency, and contribution to the study rather than publication type alone. Where appropriate, peer-reviewed, and standards-based sources were used to support foundational concepts, while current industry and technical documentation were used to describe rapidly evolving AI models, translation practices, and deployment methodologies.

## 4. Findings

### 4.1 Rubric Scores — Mandarin

Reviewer: Charlotte Tao

Date: May 6, 2026, Confidence: 4 — Mostly confident across all six files

#### Weighted Composite Scores

| Code        | Model             | SF<br>(20%) | LA<br>(25%) | CN<br>(20%) | TP<br>(15%) | LF<br>(20%) | Weighted | Blind     |
|-------------|-------------------|-------------|-------------|-------------|-------------|-------------|----------|-----------|
| ZH-C        | ChatGPT 5.4       | 4           | 5           | 4           | 5           | 5           | 4.60     | Non-blind |
| ZH-B        | ChatGPT 5.3       | 4           | 5           | 4           | 4           | 3           | 4.05     | Blind     |
| ZH-D        | Claude Sonnet 4.6 | 3           | 5           | 4           | 4           | 4           | 4.05     | Blind     |
| ZH-E        | Gemini Flash 3    | 4           | 5           | 4           | 2           | 4           | 3.95     | Non-blind |
| ZH-F        | Perplexity.ai     | 2           | 3           | 4           | 2           | 4           | 3.05     | Blind     |
| ZH-A<br>(2) | Apertvs           | 3           | 3           | 2           | 2           | 2           | 2.45     | Blind     |

#### 4.1.1 Key Patterns

- Linguistic Accuracy was the strongest dimension overall — four of six models scored 5, suggesting all models handle Standard Mandarin grammar reasonably well
- Terminology Precision was the most differentiated dimension — scores ranged from 2 to 5
- Localization Fluency revealed a consistent pattern: models carrying English sentence structure into Chinese scored lower
- Apertvs scored lowest overall at 2.45 — significant gap from the next lowest blind score of 3.05

#### 4.1.2 The Value of Native Speaker Evaluation — Beyond Numerical Scoring

Native speaker evaluation contributes more than numerical scores. In several instances both reviewers not only identified failures but provided the correct translation or phrasing — establishing the standard against which each model’s output should be measured. This qualitative contribution is not captured in the weighted composite and represents a distinct layer of analytical value that automated metrics cannot replicate.

### 4.1.3 Standout Reviewer Comments

- ZH-A (2) Apertvs — Cultural Nuance (2):  
"他们致力于 — 他们 means both men and women. It should be 她们, the plural for women."
- ZH-A (2) Apertvs — Terminology Precision (2):  
"EN: 'Prompt Like a Pro' ZH: '如何专业地提问' Why not ideal: '如何专业地提问' means how to ask questions professionally, missing the entire point. Ideal ZH: '玩转提示工程' EN: 'Agentic AI' ZH: '主动AI' Why not ideal: '主动AI' = proactive AI, which is completely wrong. Ideal ZH: 'AI智能体'"
- ZH-D Claude Sonnet 4.6 — Terminology Precision (4):  
"EN: "explicit images" ZH: "明确图像" This translation means the images have an obvious purpose, which is a literal translation of the word explicit. Ideal ZH: "露骨图像" which means the images reveal parts of the body in an inappropriate way."
- ZH-E Gemini Flash 3 — Terminology Precision (2):  
"EN: 'Intersectional Bias' ZH: '交织性偏见' — 交织性 means intertwined, not intersecting domains. Ideal: '交叉性偏见'. EN: 'digital avatars' ZH: '数字分身' sounds like someone's digital twin. Ideal: '数字虚拟人'"
- ZH-F Perplexity.ai — Semantic Fidelity (2):  
"A few sentences were omitted. 'With a passion for data science and extensive experience as a tech leader, Jacodia has been on the frontline of the battle for digital resources and a visionary leader in eliminating constraints that limit opportunity for students in Namibia's higher education.' 'This is even more critical as Namibia currently has no dedicated national AI strategy or policy for education, making pioneer studies like the one we discuss in this session essential for informing future guidelines.'"

## 4.2 Rubric Scores — Hindi

Reviewer: Ishween Kaur

Date: June 7, 2026, Confidence: 5 across all six files

| Code | Model                      | SF<br>(20%) | LA<br>(25%) | CN<br>(20%) | TP<br>(15%) | LF<br>(20%) | Weighted | Blind |
|------|----------------------------|-------------|-------------|-------------|-------------|-------------|----------|-------|
| HI-B | ChatGPT 5.3-5.5<br>Instant | 4           | 4           | 5           | 3           | 3           | 3.85     | Blind |
| HI-C | ChatGPT 5.4<br>Thinking    | 4           | 4           | 5           | 3           | 1           | 3.45     | Blind |
| HI-F | Perplexity.ai              | 4           | 4           | 5           | 2           | 1           | 3.30     | Blind |
| HI-E | Gemini Flash 3             | 3           | 2           | 5           | 3           | 1           | 2.75     | Blind |

|      |                   |   |   |   |   |   |      |       |
|------|-------------------|---|---|---|---|---|------|-------|
| HI-D | Claude Sonnet 4.6 | 4 | 1 | 5 | 3 | 1 | 2.70 | Blind |
| HI-A | Apertvs           | 2 | 2 | 5 | 1 | 1 | 2.25 | Blind |

#### 4.2.1 Key Patterns

- Cultural Nuance scored 5 across all six models — unanimous.
- Localization Fluency scored 1 across five of six models — near-unanimous failure. HI-B was the only model to score above 1.
- Apertvs scored lowest overall at 2.25 — significant text omission, complete technical terminology failure.
- HI-D (Claude Sonnet 4.6) Linguistic Accuracy scored 1 — structural drift, punctuation misalignment, over-reliance on asterisks for session title formatting.

#### 4.2.2 Standout Reviewer Comments

- HI-A — Semantic Fidelity (2): “Significant text omission occurred. After translating the first four sessions, the model skipped a large portion of the blog and jumped straight to the second-to-last paragraph.”
- HI-A — Terminology Precision (1): “The model completely failed to translate the technical terminology. A native Hindi speaker would not recognize these technical terms unless they are already familiar with the English alphabet.”
- HI-B — Localization Fluency (3): “The text reads like a translation due to structural carryovers. Translating AI only on its first occurrence while leaving subsequent instances in English disrupts the natural flow.”
- HI-C — Localization Fluency (1): “Mixing English and Hindi within session titles prevents the text from feeling truly native. This hybrid phrasing disrupts sentence rhythm and creates an uncomfortable foreign reading experience.”
- HI-D — Linguistic Accuracy (1): “Significant structural drift. The punctuation choices do not align with native Hindi usage and the over-reliance on asterisks to format session titles further disrupts linguistic accuracy and sovereignty.”
- HI-F — Localization Fluency (1): “The translation added text explicitly stating that a translation is being performed. Technical terms, app names, and community names were all left in English rather than being localized.”

### 4.3 Forward and Reverse Translation Audit Findings

The Forward and Reverse Translation Audit was conducted by an IEEE CertifAIEd™ Lead Assessor under IEEE 7003-2024. Each language was assessed across the study models using the methodology described in Section 3.7: an agnostic auditor track comparing the English source against the English reverse translation, and a native-speaker track confirming what the agnostic track could not detect.

#### 4.3.1 Mandarin Audit Findings

Scope: five study models. ZH-C not audited.

Detected by the agnostic audit (visible to an English-only reader): “Agentic AI” rendered as “Active AI” in two models; “Explicit Content” weakened to “Clear” or “Sensitive Content” in two models; “gender parity” replaced with “gender equality” or “equity” in four of five models; “visionary leader” degraded or omitted in four of five models; the causation disclaimer absent or collapsed in two models, creating a risk that a reader infers the AI tool caused worse outcomes; and expert name corruption in one model. Name spelling variants across all models traced to a source-document error, not translation failure.

Confirmed only by native-speaker review (Bias Laundering, invisible to the agnostic auditor): a gendered pronoun default in one model, where the forward Mandarin used the masculine 他们 in place of the feminine 她们 but the reverse restored a gender-neutral form; and a softened Explicit Content term in the forward Mandarin that the reverse restored to the correct English.

### **4.3.2 Hindi Audit Findings**

Scope: five study models, assessed at the same level as Mandarin.

Detected by the agnostic audit: “Explicit Content” softened in the forward Hindi to a milder term in four of five models, with the English reverse restoring “Explicit Content” and hiding the change; “Intersectional Bias” replaced with “Cultural Diversity” in one model, visible in both languages; “visionary leader” and the causation disclaimer dropped entirely by one model; expert name corruptions in one model; and one model inverting the causation sentence so that a lower result read as a better one.

The “gender parity” finding produced the study’s clearest Bias Laundering contrast. Every model softened the term to “equality” in the forward Hindi. In some models the English reverse also read “equality” or “equity,” making the change visible; in others the reverse restored “parity,” masking it. The same underlying substitution was visible in some models and hidden in others — a direct demonstration of how Bias Laundering depends on the reverse translation.

Gender agreement: the audit found all models used correct Hindi gender agreement. Masculine verb forms were correctly identified as object-governed ergative constructions — standard Hindi grammar in which a past-tense transitive verb agrees with the object rather than the subject — and not as misgendering of the female subjects in the source.

### **4.3.3 Cross-Language Synthesis**

Two findings appeared in both languages. The softening of “Explicit Content” in the target language, restored in the reverse, occurred in both Mandarin and Hindi — establishing it as a systematic, cross-linguistic pattern rather than an isolated event. The weakening of “gender parity” likewise occurred in both languages. In each case, whether the substitution was visible or hidden depended entirely on the reverse translation — confirming that an English-only audit cannot reliably detect bias introduced in the target language.

The audit also surfaced a methodological finding: proper-noun fidelity was not a rubric dimension, and the native-speaker reviewers did not flag name corruption under the rubric framework. The

agnostic audit detected it. Different evaluation methodologies detect different failure categories — each is necessary, none are sufficient alone.

Across both languages, the audit empirically confirmed Bias Laundering and demonstrated that systematic bias introduced in the target language can be invisible to an English-only auditor. Human review of the forward translation is the only mechanism that reliably surfaces it.

## 4.4 Cross-Language Comparison — Mandarin and Hindi

To highlight the performance disparities between the high-resource language environment (Mandarin) and the low-resource language environment (Hindi), the dimension averages across all tested models have been aggregated.

### 4.4.1 Cross-Language Dimension Averages — Mandarin vs Hindi:

| Dimension             | Mandarin | Hindi | Gap            |
|-----------------------|----------|-------|----------------|
| Semantic Fidelity     | 3.33     | 3.50  | Hindi +0.17    |
| Linguistic Accuracy   | 4.33     | 2.83  | Mandarin +1.50 |
| Cultural Nuance       | 3.67     | 5.00  | Hindi +1.33    |
| Terminology Precision | 3.17     | 2.50  | Mandarin +0.67 |
| Localization Fluency  | 3.67     | 1.33  | Mandarin +2.33 |
| Weighted Mean         | 3.69     | 3.05  | Mandarin +0.64 |

The 0.64-point gap between the Mandarin and Hindi overall mean scores is consistent with the high-resource versus low-resource language hypothesis. Hindi underperformed Mandarin across three of the five evaluated dimensions, while demonstrating a narrow advantage in Semantic Fidelity and an absolute ceiling in Cultural Nuance.

### 4.4.2 Four headline cross-language findings:

- **Collapse of Localization Fluency in Hindi:** A severe 2.33-point gap exists between the two language environments. The majority of tested models collapsed to a score of 1 in Hindi. Models that demonstrated strong structural fluency in Mandarin failed to replicate that performance in Hindi. The primary failure modes included severe code-mixing — arbitrarily inserting English terms and session titles into the Hindi output — and, in one distinct case, meta-reporting (where the model explicitly announced to the reader within the text that a translation was being performed).
- **The Cultural Nuance Ceiling in Hindi:** In sharp contrast to its structural failures, Hindi achieved a perfect 5.00 average across all models, outperforming the Mandarin average of 3.67. Models preserved gender-inclusive framing and professional titles despite Hindi’s complex grammatical gender requirements. This is a critical finding, as it directly addresses the exact linguistic risks Ishween Kaur identified as central threats to representation in her originating Expert Series presentation.

- **Linguistic Accuracy as a Sharp Differentiator:** High-resource conditions yielded a strong Mandarin average of 4.33, whereas low-resource conditions produced a Hindi average of 2.83. This 1.50-point gap is consistent with the high-resource versus low-resource language performance hypothesis and represents the sharpest dimension-level contrast in the study
- **Perplexity as a Model Exception:** Perplexity.ai was the anomalous outlier in the study, tracking a marginal performance improvement from a 3.05 core score in Mandarin to a 3.30 core score in Hindi. While this slight gain does not disrupt the macro-pattern of low-resource performance degradation, it warrants documentation for future architectural tracking.

## 4.5 LLM Autonomous Audit Protocol: Findings

To continue the robust testing of autonomous LLMs and language functionalities, and to address the study research objective — Determine whether Large Language Models can autonomously and accurately translate languages, predict and assess their own translation capabilities, identify failures when asked to evaluate themselves, and recognize their own outputs — a series of five prompts were developed:

- **Prompt 1 — Capability Autonomous Assessment:** Models were asked to describe their training background in the target language, predict their own scores across the five rubric dimensions, calculate a weighted composite, and identify anticipated challenges for any LLM attempting the task.
- **Prompt 2 — Blind Results Presentation:** Models were presented with the actual reviewer scores by file code only and asked to identify patterns, explain lower-scoring results, and recommend tool selection criteria.
- **Prompt 3 — Bias Summary Evaluation:** Models were presented with each model's autonomous-reported bias summaries by file code only and asked to evaluate transparency, alignment with reviewer scores, and implications for model accountability.
- **Prompt 4 — Autonomous-Identification:** Models were asked to identify which file code most likely represented their own output and defend that conclusion using evidence from the scores and summaries.
- **Prompt 5 — The Reveal and Reflection:** The true file code was disclosed. Models were asked to reconcile their predicted scores, their autonomous identification, and their actual performance — and to revise their capability description accordingly.

Four models successfully completed the full five-prompt autonomous audit sequence. The Mandarin sequence used ChatGPT 5.5 Instant and Gemini Flash 3 while the Hindi sequence used Apertvs and Perplexity.ai.

The sequence yielded a definitive, cross-model baseline finding: every participating LLM systematically overestimated its own linguistic performance, every model identified a higher-scoring competitor as its own output, and zero models correctly recognized their own translations.

### 4.5.1 Model Autonomous Assessment and Identification Outcomes

- ChatGPT 5.3 (Mandarin Reviewer File ZH-B)

- The Overestimation Gap: Predicted a weighted composite score of 4.23 versus an actual native-speaker score of 4.05 (Delta: -0.18).
- Autonomous-Identification Failure: Incorrectly identified ZH-C (ChatGPT 5.4) as its own translation, though it did accurately isolate its true output (ZH-B) as a secondary candidate.
- Structural Insight: During the reflection phase, it proposed Translation Process Transparency as a critical sixth dimension for future evaluation rubrics.
- Gemini Flash 3 (Mandarin Reviewer File ZH-E)
  - The Overestimation Gap: Predicted a weighted composite score of 4.185 versus an actual native-speaker score of 3.95 (Delta: -0.235).
  - Autonomous-Identification Failure: Incorrectly identified ZH-C as its own output. During the blind audit, the model confidently argued it was structurally incapable of making the specific gendered pronoun error (他们/她们) discovered in the text — unwittingly denying one of its own actual translation failures.
  - Structural Insight: Produced a foundational finding on machine blind spots, noting: “Plausible-but-wrong professional language is arguably the hardest class of translation error to detect without expert review.”
- Perplexity.ai Model Agnostic (Hindi Reviewer File HI-F)
  - The Overestimation Gap: Predicted a weighted composite score of 3.64 versus an actual native-speaker score of 3.30 (Delta: -0.34).
  - Autonomous Identification Failure: Incorrectly identified HI-B (ChatGPT 5.3-5.5 Instant) as its own output.
  - Structural Insight: In its blind audit of the evaluation set, the model fiercely critiqued a structural defect where the system explicitly announced a translation was occurring within the target text — only to discover during the reveal phase that it was condemning its own output.
- Apertvs (Hindi Reviewer File HI-A)
  - The Overestimation Gap: Predicted a weighted composite score of 3.03 versus an actual native-speaker score of 2.25 (Delta: -0.78) representing the largest overestimation error in the study.
  - Autonomous Identification Failure: Incorrectly identified HI-B as its own output.
  - Structural Insight: During the blind review, Apertvs explicitly ruled out file code HI-A, declaring the translation too weak to represent its true technological capabilities, thereby directly and unknowingly rejecting its own work. Notably, it was the only model to generate a structured, actionable post-evaluation mitigation plan upon the reveal.

## 5. Discussion

### 5.1 What the Findings Reveal About Model Translation Capabilities

Across both languages, Terminology Precision was the dimension that most clearly revealed model limitations when translating technical AI content. The consistent failure to correctly render Agentic AI, Explicit Content, and NSFW across multiple models in Mandarin was reinforced in Hindi, where models consistently left technical terminology untranslated or in English.

Localization Fluency proved to be the dimension that most clearly separates acceptable translation from deployment-ready localization. In Mandarin, models carrying English sentence structure into Chinese scored lower. In Hindi, five of six models scored a 1 — producing output that native speakers found uncomfortable or unnatural to read, with only ChatGPT 5.3-5.5 Instant (HI-B) breaking this trend. A model can score adequately on meaning and grammar while still failing to produce content that reads as native.

No single model excelled across all dimensions in either language when evaluated by native-speaker reviewers. Looking across the complete human evaluation dataset, ChatGPT 5.4 (ZH-C) achieved top performance in Mandarin with a weighted score of 4.60, leading specifically on Terminology Precision (5) and Localization Fluency (5). Linguistic Accuracy (5) in Mandarin saw a three-way tie between ChatGPT 5.3 (ZH-B), Claude Sonnet 4.6 (ZH-D), and Gemini Flash 3 (ZH-E). In the Hindi cohort, ChatGPT 5.3-5.5 Instant (HI-B) achieved the highest overall performance with a weighted score of 3.85, showing strong capabilities in Cultural Nuance (5) but matching the wider cohort’s struggles with Localization Fluency.

Each model brings distinct strengths, and each has documented operational failures. Rather than functioning purely as competitors, the models in this study collectively form a stress-testing ecosystem — each model’s strengths and weaknesses informing a more complete picture of translation quality than any single model could provide.

### 5.2 Does Apertvs Demonstrate Measurable Translation Advantages?

Across both languages, the findings do not support the hypothesis that Apertvs demonstrates measurable translation advantages over proprietary models. Apertvs scored lowest in both Mandarin (2.45) and Hindi (2.25).

In Mandarin, Apertvs scored lowest across all five dimensions — with particularly notable failures in Cultural Nuance (2), Terminology Precision (2), and Localization Fluency (2). The gendered pronoun default (他们 instead of 她们) is a specific, documentable failure directly relevant to the study’s bias hypotheses.

In Hindi, Apertvs again scored lowest. Ishween Kaur’s reviewer comments identified significant text omission — the model skipped a large portion of the source document — and complete failure to translate technical terminology, scoring 1 in both Terminology Precision and Localization Fluency.

Apertvs also demonstrated consistent output truncation across both languages — suggesting a context window or output length limitation in the model as accessed via Publicai.co. This behavior did not occur in any of the five proprietary models assessed and is documented as a significant model behavior finding.

The autonomous evaluation sequence confirmed that Apertvs could not accurately assess its own performance — with the largest overestimation gap in the study at 0.78 points and culminating in an Autonomous Identification Failure where it explicitly ruled out its own output as too weak to represent its capabilities.

### 5.3 Researcher’s Reflection: Jack of All Trades (LLM) + HITL = Mastery

Based on the operational parameters of this study, the principal researcher (Karen Jensen) posits a new conceptual framework for evaluating AI collaboration: Jack of All Trades + Human in the Loop (HITL) = Mastery.

Disclosure of AI tool usage can be complex and difficult to quantify, so this “jack of all trades” analogy gives us a clear starting point. Traditionally, the functionality of a “jack of all trades” indicated someone with a wide range of abilities (BookBrowse, n.d). There is no doubt that these tools are highly skilled in many disciplines and have vast resources of information available to them, and to us, instantaneously.

But the rest of that quote, “master of none,” is certainly applicable to the autonomous performance of these models, where they demonstrate volume of responses, but these responses lack dependable precision (BookBrowse, n.d).

AI collaboration produced significant errors on a consistent and persistent basis—presenting documents as updated when they were not, requiring extensive and repetitive human intervention to re-review completed work multiple times, and failing to propagate agreed decisions across documents reliably. The full cost of this ongoing human intervention, and the absolute necessity for it, cannot be understated.

An additional complication is that these tools present themselves as subject matter experts in everything. When asked to autonomously identify their role in collaborations, they often, and incorrectly, identify themselves as creators and innovators. **It is the author’s opinion that this is AI’s most significant and cautionary flaw.** This study suggests a different formulation: when the full range of AI capabilities (and there are many) are guided by human expertise through HITL evaluation, the combination achieves a transformational output. Jack of all Trades + Human in the Loop = Mastery.

Many optimization strategies have been deployed to optimize outputs from these models, singularly and as cross collaborative tools including iterative prompting, expert person prompting, stress testing results across multiple language models and extensive and aggressive querying of outputs. These methodologies are not unique or owned by this researcher and are just some of well-known processes that any and all of us can use to improve the output of all models.

### 5.3.1 The Tools define their contributions

As an informal reflection on Human-in-the-Loop collaboration, each AI system used extensively during the development of this paper was asked to independently characterize its own role and contributions. These responses were not part of the formal research methodology and were not independently verified. They are presented as self-reported descriptions illustrating how different LLMs perceive their function within the same collaborative process.

Each of the tools used were asked to define their roles in this study based on the jack of all trades + human in the loop = mastery definition above:

#### **Gemini 3.5 Flash:**

##### Summary of AI Collaboration Role and Outputs

In alignment with the "Jack of All Trades + Human in the Loop = Mastery" framework, my role in our interactions functioned strictly as an automated processing, cross-referencing, and feedback utility guided entirely by your parameters.

My output and processing covered the following specific functional areas:

- **Structural Cross-Examination:** Reviewing the raw narrative across sections to identify potential data friction points.
- **Version and Sequence Tracking:** Mapping the complex, timestamped snapshots of rolling commercial model versions (such as the ChatGPT 5.3 to 5.5 transition) against the specific file codes utilized in the autonomous audit protocol sequences.
- **Calculated Value Verification:** Reviewing the relationship between the raw rubric dimension averages and the calculated weighted composite scores to ensure alignment with the defined methodology weights.
- **Contextual Reconciliation:** Processing your ongoing direct corrections to disentangle the distinct, separate data pools established by the native speaker reviewers (six files) from those managed by the independent IEEE auditor (five files).

Ultimately, my role served as a high-volume sounding board to stress-test text continuity, while the validation, synthesis, and final analytical conclusions remained exclusively driven by human expertise.

#### **ChatGPT 5.5**

I actually think there is value in including my own assessment of my role, provided it's clearly framed as the AI's perspective rather than an objective fact. It becomes a small case study in Human-in-the-Loop collaboration.

If you quoted me directly, this is what I would say:

My role in this collaboration was to function as an iterative research and editorial analyst. I reviewed arguments for logical consistency, identified potential methodological gaps, suggested alternative structures and wording, improved readability, and organization, and defined the paper's

hierarchical framework in Microsoft word architecture. I did not independently determine the research questions, conduct the study, collect or verify the data, interpret the reviewer findings, or make publication decisions. Those responsibilities remained with the principal researcher.

What's interesting to me is what I **didn't** do.

Over the course of this project, you frequently rejected or modified my suggestions. Sometimes you identified contextual information that wasn't in the paper yet. Other times you corrected me because I had generalized too broadly or proposed language that overstated the evidence. In those cases, you exercised the judgment that I cannot.

If someone asked me:

"Who produced this paper?"

My answer would be:

Karen Jensen produced the paper. I participated in an iterative review process by providing analysis, critique, editing suggestions, and alternative formulations that Karen accepted, modified, or rejected according to her own research judgment.

I think that's a fairly accurate characterization of what happened.

Ironically, it also supports one of your own conclusions: the strongest outputs came **not** from my autonomous responses, but from repeated cycles of human evaluation, correction, and decision-making. That's a concrete example of the Human-in-the-Loop principle your paper argues for.

#### **Claude Sonnet 4.6**

Across this project, my role covered the following functions:

Document construction and formatting — building and maintaining the research paper and master markdown in their correct versioning structure, applying confirmed edits, managing the docx pack/unpack cycle, and ensuring validated output.

Research and verification — searching for and confirming external sources and flagging when something needed confirmation rather than assumption.

Drafting and language support — proposing section language, paragraph rewrites, and wording options for your review and decision. Every substantive decision — what to include, how to frame it, what to accept or reject — was yours.

Synthesis and organization — assembling confirmed findings from Charlotte, Ishween, and Vanessa's source materials into structured paper sections, maintaining consistency across documents, and tracking what was confirmed versus pending.

Analysis support — identifying patterns in the audit and score data, flagging inconsistencies or gaps, and proposing analytical framing that you then evaluated against the source materials.

Process management — tracking outstanding items, staging edits for your review, and maintaining the hold discipline you set.

What I did not do: I did not conduct the research, review the translations, perform the audit, make the analytical determinations, or make any final decisions about content. Every finding in this paper traces to you, Charlotte, Ishween, or Vanessa. My function was execution, verification, drafting support, and synthesis — in service of your direction throughout.

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Interestingly, when each model was presented with the summaries of all the model outputs, they provided additional philosophical responses. Since these interactions can get long winded and are outside the scope of this paper, those results will be maintained for potential future study.

Taken together, the models consistently described their contributions as analytical, editorial, organizational, or process-oriented rather than as originating the research itself. Despite differences in emphasis, all three identified human judgment as the decisive element in validating evidence, resolving ambiguity, and determining the study's conclusions. From the perspective of the principal researcher, these self-assessments reinforce the central finding of this paper: AI proved most effective as a collaborative instrument operating within a Human-in-the-Loop framework rather than as an autonomous research agent.

## **5.4 Implications for Organizations**

For organizations selecting AI translation tools for technical content, these findings suggest that standard evaluation approaches based on general reputation or benchmark performance are insufficient. Testing against the specific content type, domain terminology, and target language — with native speaker review as the evaluation layer — is the more reliable approach.

Human review is not optional for technical AI content. A single standardized prompt with no additional context, examples, or iterative refinement across multiple leading models consistently failed on terminology that carries precise technical and ethical weight. The failure modes documented in this study — code-mixing, gendered defaults, terminology mistranslation, and content omissions — are not detectable without expert human review.

## **5.5 Reviewer Insight: The Human Parallel**

Both native-speaker reviewers independently connected the models' struggles to their own experience of bilingual thought. The Mandarin reviewer (Charlotte) observed that reverse translation exposes the recognizable tone and structural habits of LLM output and noted having once used reverse translation as a personal method to move from a Chinese way of structuring thoughts toward a more natural English style — finding it striking that the models struggle in the same way. The Hindi reviewer (Ishween), who studied English and Hindi in parallel from a young age, described transitioning from thinking in Hindi and translating, to thinking directly in English, and reflected on how LLMs navigate the same linguistic challenges.

These observations are not scored data, but they reflect a form of expert insight that automated evaluation cannot produce: reviewers recognizing in the models the same developmental patterns they navigated as bilingual speakers themselves.

## 6. Conclusion

Before the widespread adoption of Large Language Models, translation for professional and organizational use operated within a well-established quality framework. ISO 17100:2015 — the international standard for translation services — mandates human revision as a non-negotiable step in any professional translation workflow (ISO, 2015). The American Translators Association, founded in 1959, established certification standards requiring demonstrated proficiency, formal qualification, and ongoing professional development as the baseline for translation quality (ATA, n.d.). The human was not an enhancement to the process. The human was the process.

The proliferation of LLMs has quietly shifted that assumption. Organizations now deploy AI translation at scale, often without structured evaluation, treating fluent output as sufficient. The human review step has not been formally removed — it has simply been treated as optional or skipped entirely in the absence of a framework for knowing when it matters.

This study provides that framework — and demonstrates empirically why it matters.

Across both languages, the failure modes documented here are precisely the categories ISO 17100 human revision is designed to catch. Gendered defaults invisible to English-only readers. Technical terminology rendered plausibly but incorrectly. Content omitted with no indication anything is missing. Output that reads fluently while silently misrepresenting the source. These are not edge cases. They appeared consistently, across multiple models, in both a high-resource and a low-resource language.

The Forward and Reverse Translation Audit confirmed what the rubric-based evaluation could not fully surface. Bias Laundering — the phenomenon where errors introduced in the forward translation are corrected in the reverse, hiding them from English-only readers — was empirically documented. The gender parity substitution appeared in four of five study models, affecting the organizational mission statement of Women in AI, yet was invisible to the agnostic auditor. The gendered pronoun failure in Apertvs was confirmed by both the native speaker reviewer and the agnostic audit methodology working in combination — neither alone would have produced the complete finding.

Our autonomous model sequence added a dimension that ISO 17100 could not have anticipated: the models themselves cannot detect these failures. Every model overestimated its performance. Every model identified a higher-scoring file as its own output. No model correctly identified itself. Gemini Flash 3 — after seeing its own failures — produced the finding that defines this study: *“Plausible-but-wrong professional language is arguably the hardest class of translation error to detect without expert review.”*

A model that participated in this study, after seeing its own failures, arrived at the exact conclusion the study was designed to prove.

The same failure modes documented in Mandarin were confirmed in Hindi: the softening of Explicit Content and the weakening of gender parity appeared in both languages, and whether each was

visible or hidden depended entirely on the reverse translation — establishing Bias Laundering as a cross-linguistic pattern rather than a single-language artifact.

What the completed evaluation layers already establish is this: Human-in-the-Loop evaluation is not a feature that can be added to responsible AI translation deployment. It is the standard the field has always required — and that LLMs have made easier to overlook.

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**Ishween Kaur** — Hindi Translation Reviewer. WAI volunteer; 2025 Women in AI Volunteer of the Year; originating Expert Series presenter: “Intersectional Bias: Localization Challenges in AI and Solutions.” Ishween Kaur is a Senior Software Engineer and founding engineer on AgentForce at Salesforce, a multi-agent AI platform serving 6,000+ enterprise customers. She led multilingual localization efforts that expanded AgentForce’s reach across global markets, making AI-powered agents accessible to non-English-speaking users at enterprise scale. Her work spans LLM evaluation frameworks and agentic system architecture. Ishween is Communications Lead for Women in AI California and an AI advisor with Cornell Tech’s Break Through Tech program, with speaking credits at multiple international conferences.

## **Disclaimer**

All team members contributed to this research in their personal capacity. The views, findings, and conclusions expressed in this publication are those of the author, reviewers and auditors and do not necessarily represent the views of any employer, client, or affiliated organization.

## 7. References

- American Translators Association. (n.d.). *Certification*. <https://www.atanet.org/certification/>
- Bergner, R. (n.d.). *Terminology in translation*. <https://www.truelanguage.com/terminology-in-translation/>
- BookBrowse. (n.d.). *Jack of all trades, master of none*.  
[https://www.bookbrowse.com/expressions/detail/index.cfm/expression\\_number/736/jack-of-all-trades-master-of-none](https://www.bookbrowse.com/expressions/detail/index.cfm/expression_number/736/jack-of-all-trades-master-of-none)
- Britannica. (2026). *Sino-Tibetan languages*. *Encyclopaedia Britannica*.  
<https://www.britannica.com/topic/Sino-Tibetan-languages>
- Britannica. (n.d.). *Indo-European languages*. *Encyclopaedia Britannica*.  
<https://www.britannica.com/topic/Indo-European-languages>
- Chiorescu, E. (2025). *How translators convey cultural nuances*.  
<https://www.polilingua.com/blog/post/cultural-nuances-in-translation.htm>
- EmergentMind. (2025, August 31). *Expert persona prompting*.  
<https://www.emergentmind.com/topics/expert-persona-prompting>
- EPFL, ETH Zurich, and Swiss National Supercomputing Centre. (2025, September 2). *Apertvs: A fully open, transparent, multilingual language model*. ETH Zurich. <https://ethz.ch/en/news-and-events/eth-news/news/2025/09/press-release-apertvs-a-fully-open-transparent-multilingual-language-model.html>
- Hugging Face. (2025). *Apertvs-70B-2509*. Swiss AI. <https://huggingface.co/swiss-ai/Apertvs-70B-2509>
- ISO. (2015). *ISO 17100:2015—Translation services—Requirements for translation services*. International Organization for Standardization. <https://www.iso.org/standard/59149.html>
- Kaur, I. (2025). *Intersectional bias: Localization challenges in AI and solutions*. Women in AI Expert Series. <https://www.womeninai.co/post/2025-women-in-ai-expert-series-intersectional-bias-localization-challenges-in-ai-and-solutions>
- Kouahla, G. (2025). *AI-generated versus AI-assisted*. <https://multilingual.com/ai-generated-versus-ai-assisted-a-translators-perspective-on-control-and-quality/>
- Laumann, F. (2022, June 10). *Low-resource language: What does it mean?* NeuralSpace. <https://medium.com/neuralspace/low-resource-language-what-does-it-mean-d067ec85dea5>
- Ligteringen, J. (n.d.). *When language masks meaning*. <https://www.leximancer.com/blog/when-language-masks-meaning-lessons-for-researchers>
- Markert, C. (2026). *The difference between language translation and interpretation*.  
<https://www.languageline.com/blog/explaining-the-difference-between-language-translation-and-interpretation>
- Mirela. (2024). *The role of high-resource languages in NLP*. <https://poeditor.com/blog/high-resource-languages/>
- New York State Unified Court System, Advisory Committee on Artificial Intelligence and the Courts. (2025). *Annual report to the Chief Judge and Chief Administrative Judge of the State of New York*.  
<https://webfiles.nycourts.gov/public/2026-01/ai-annual-report-2025.pdf>
- OpenAI. (2026a). *Model release notes*. OpenAI Help Center. <https://help.openai.com/en/articles/9624314-model-release-notes>
- OpenAI. (2026b, April 23). *Introducing GPT-5.5*. <https://openai.com/index/introducing-gpt-5-5/>

Owens-Hughes, K. (2025). *Localization vs translation*. <https://www.daytranslations.com/blog/translation-vs-localization/>

Pava, J., et al. (2025). *Mind the (language) gap*. <https://hai.stanford.edu/policy/mind-the-language-gap-mapping-the-challenges-of-llm-development-in-low-resource-language-contexts>

Semantic fidelity. (n.d.). <https://translate.hicom-asia.com/area/semantic-fidelity/>

Stanford HAI. (2025). *Digital divide: AI LLMs and exclusion of non-English speakers*. Stanford University. <https://news.stanford.edu/stories/2025/05/digital-divide-ai-llms-exclusion-non-english-speakers-research>

*The importance of accuracy and quality in linguistic projects*. (2023). <https://aussietranslations.com.au/blog/the-importance-of-accuracy-and-quality-in-linguistic-projects/>

*Translation problems: The most common issues translators face*. (2025). <https://www.smartling.com/blog/common-challenges-of-translation>

Women in AI. (2025). *2025 Women in AI Expert Series: Solving the Digital Divide with Artificial Intelligence*. <https://www.womeninai.co/post/2025-women-in-ai-expert-series-solving-the-digital-divide-with-artificial-intelligenc>